

Global Earthquake Analysis

Exploring significant seismic events since 1900



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Why this project?

- Earthquakes can strongly affect communities and infrastructure.
- Historical earthquake data can show where major events are concentrated and how their characteristics differ.
- This project analyzes a global database of significant seismic activity since 1900.

Problem Statement

- Earthquake activity is not distributed equally across locations or time.
- The analysis identifies where significant earthquakes are most frequent and strongest.
- It also examines changes over time and relationships between magnitude, depth, reporting stations, and event type.
- The goal is to provide clear historical patterns that can support further preparedness and infrastructure assessment.

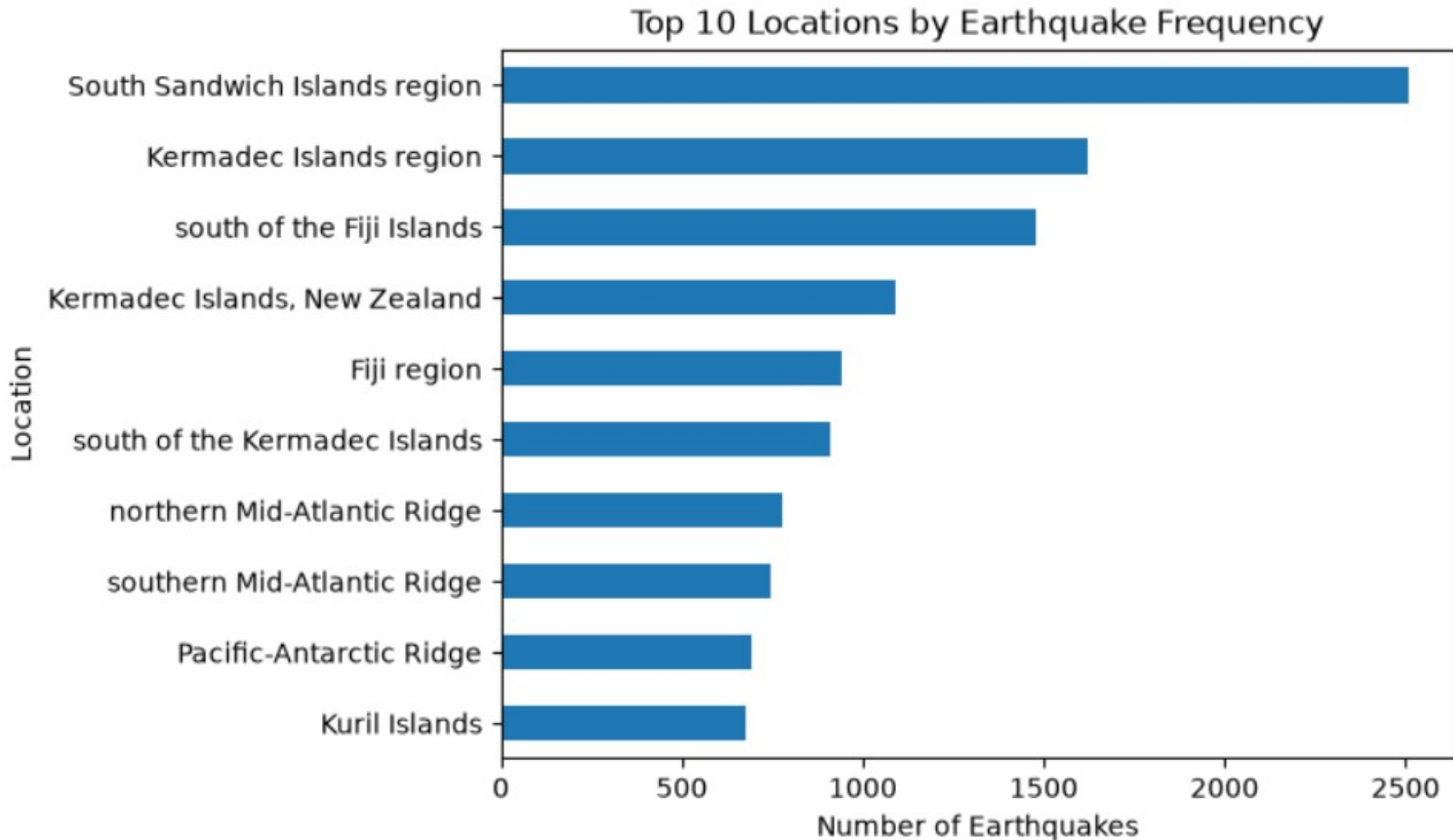
Dataset at a Glance

- Time coverage: 1900-2026
- 109,282 earthquake records after filtering to earthquake events
- 53,708 earthquake records include reporting-station (nst) data - about 49.1%
- South Sandwich Islands region has the highest recorded frequency: 2,509 events
- Source: USGS significant seismic events dataset

How to read the analysis

- Magnitude: a measure of earthquake strength.
- Depth: how far below the reference surface the event occurred, measured in km.
- Location: the dataset's place classification.
- Correlation: shows the strength and direction of a linear relationship between two numeric variables.

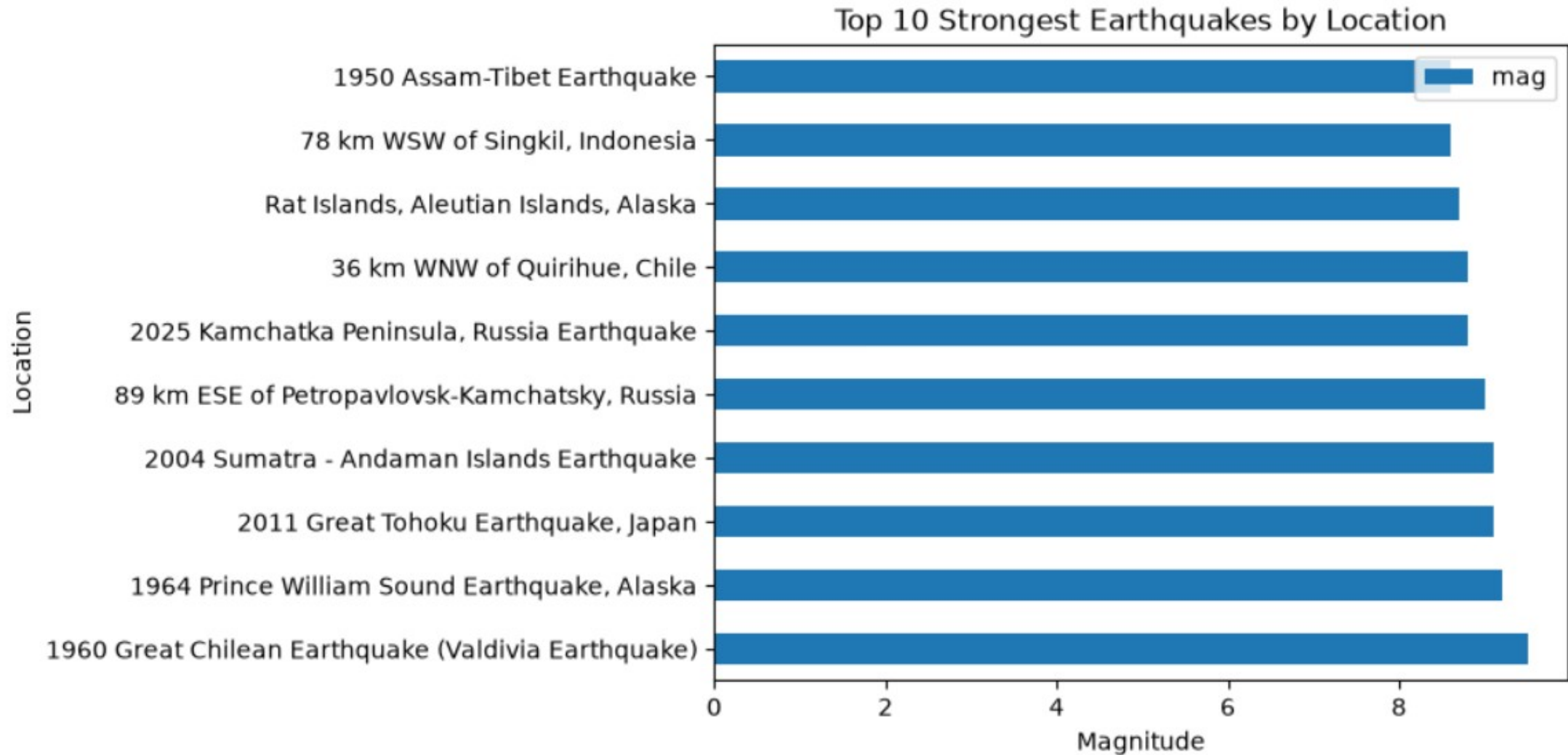
Top 10 Locations by Earthquake Frequency



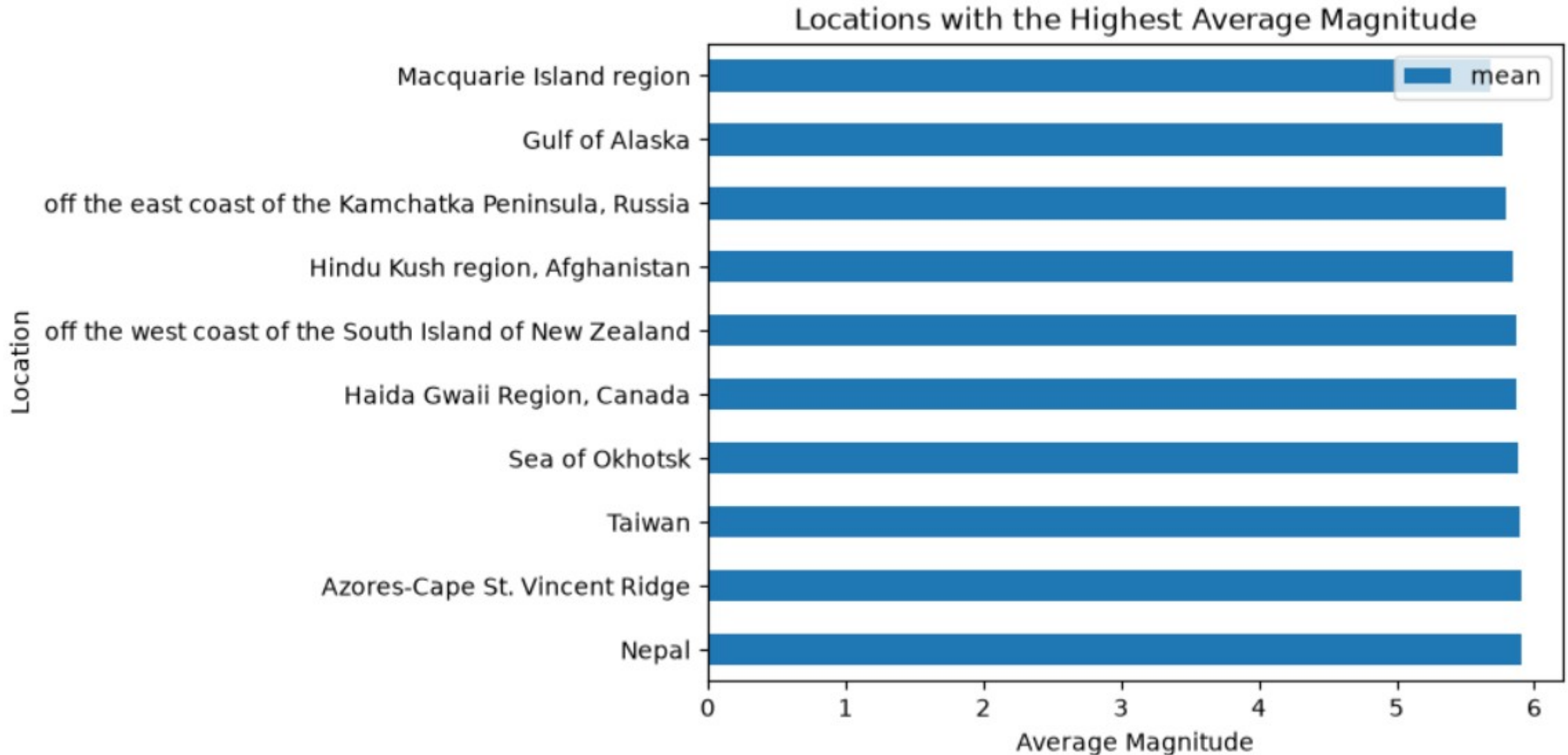
Objective 1 - Key Finding

- South Sandwich Islands region recorded the most earthquakes: 2,509.
- Kermadec Islands region followed with 1,620, and south of the Fiji Islands with 1,477.
- Recorded earthquake activity is concentrated more heavily in some locations than others.

Top 10 Strongest Earthquakes by Location



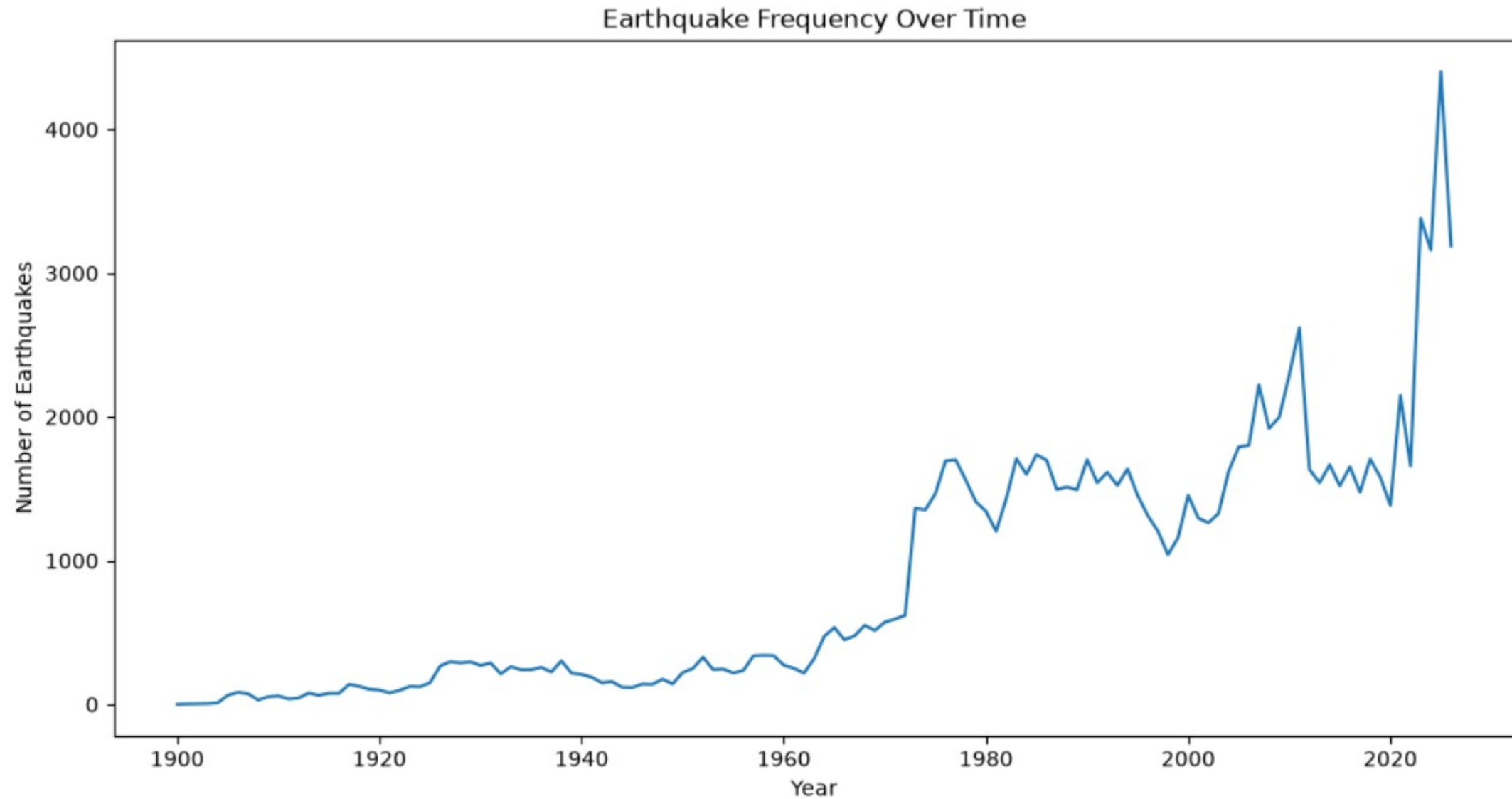
Locations with the Highest Average Magnitude



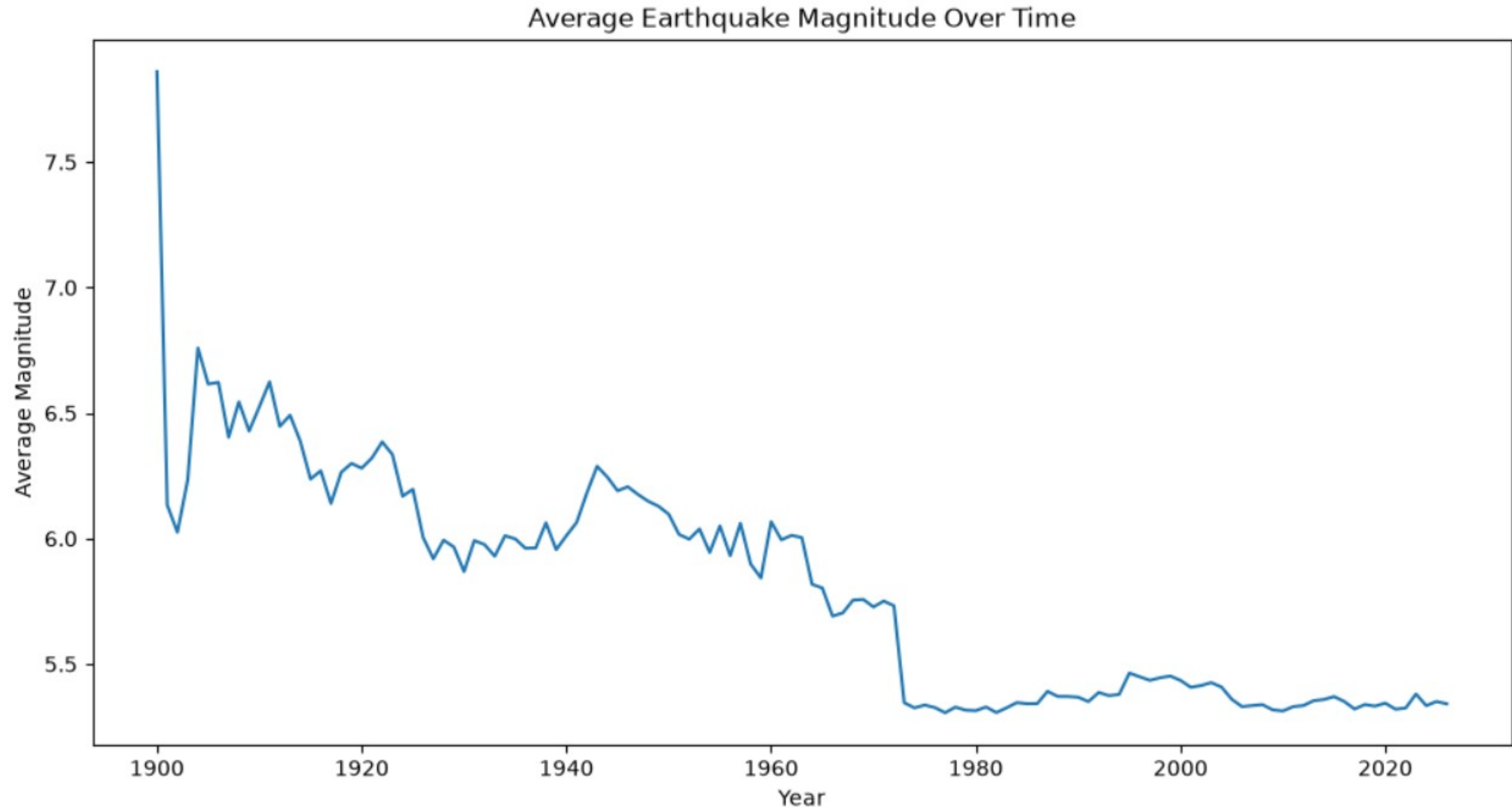
Objective 2 - Key Findings

- The strongest event was the 1960 Great Chilean (Valdivia) Earthquake at magnitude 9.5.
- For average magnitude, only locations with at least 10 earthquake records were compared.
- Nepal had the highest average magnitude at about 5.91, followed closely by Azores-Cape St. Vincent Ridge and Taiwan.

Earthquake Frequency Over Time



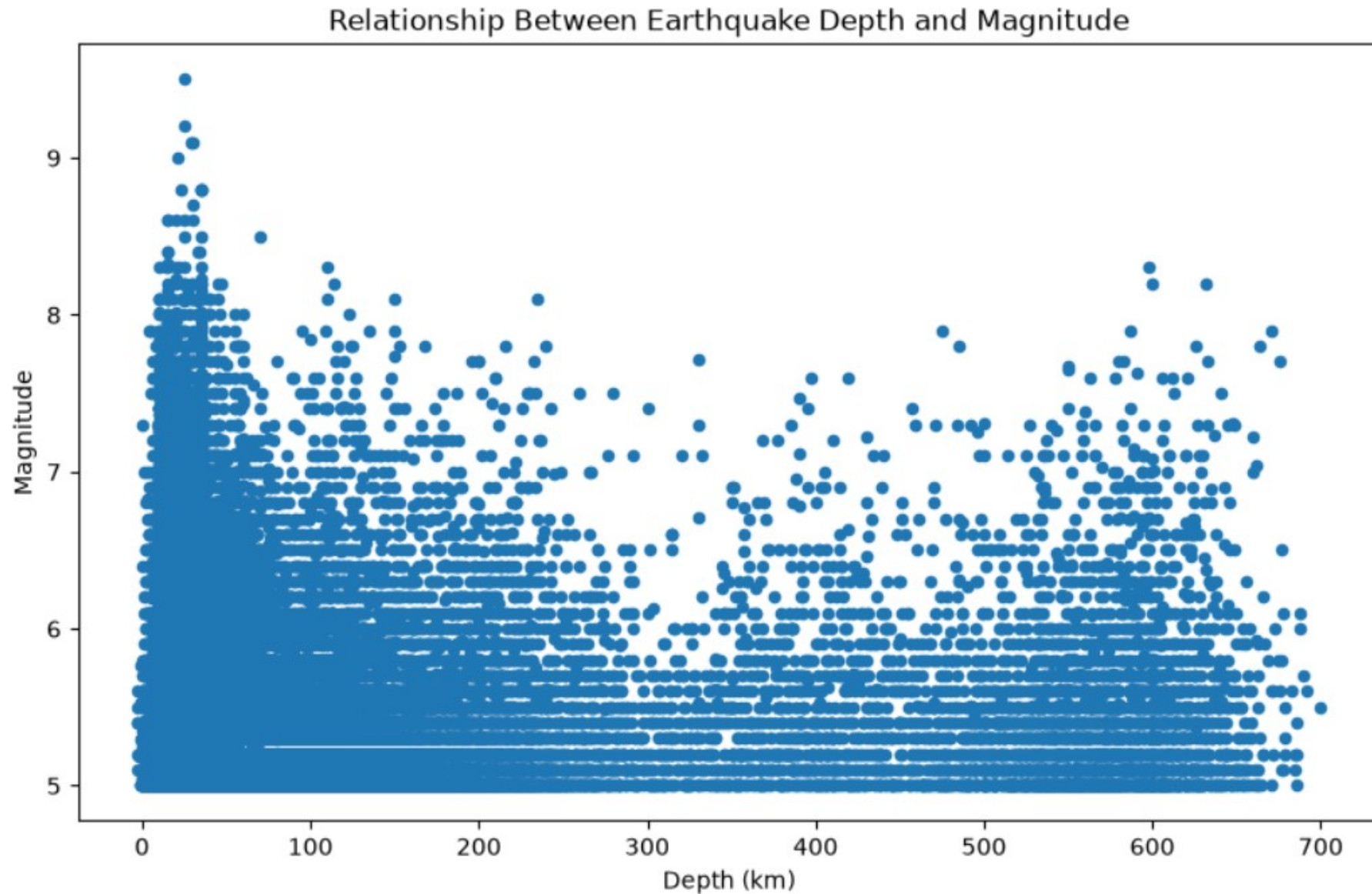
Average Earthquake Magnitude Over Time



Objective 3 - Key Findings

- Recorded earthquake frequency generally increased over time; 2025 had 4,401 recorded events.
- Average recorded magnitude decreased over the long term and has stayed around 5.3-5.4 in recent decades.
- The rise in recorded frequency may partly reflect improvements in detection and reporting, not only physical change.

Relationship Between Earthquake Depth and Magnitude

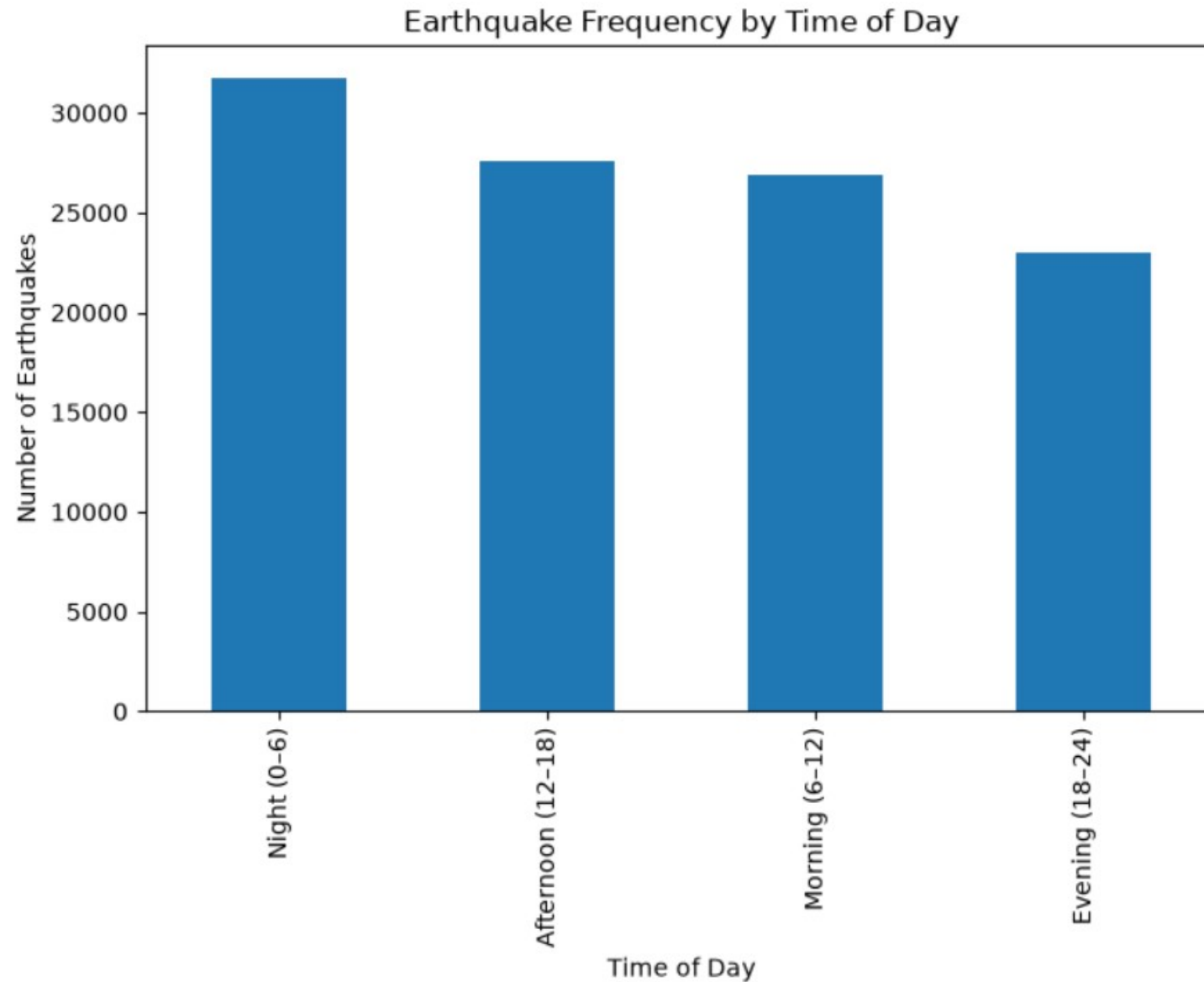


Objective 4 - Key Finding

- The correlation between depth and magnitude is about -0.01.
- This indicates almost no linear relationship in this dataset.
- Deeper earthquakes are not necessarily stronger or weaker based on these records.

Why this plot? To check whether depth and magnitude move together.

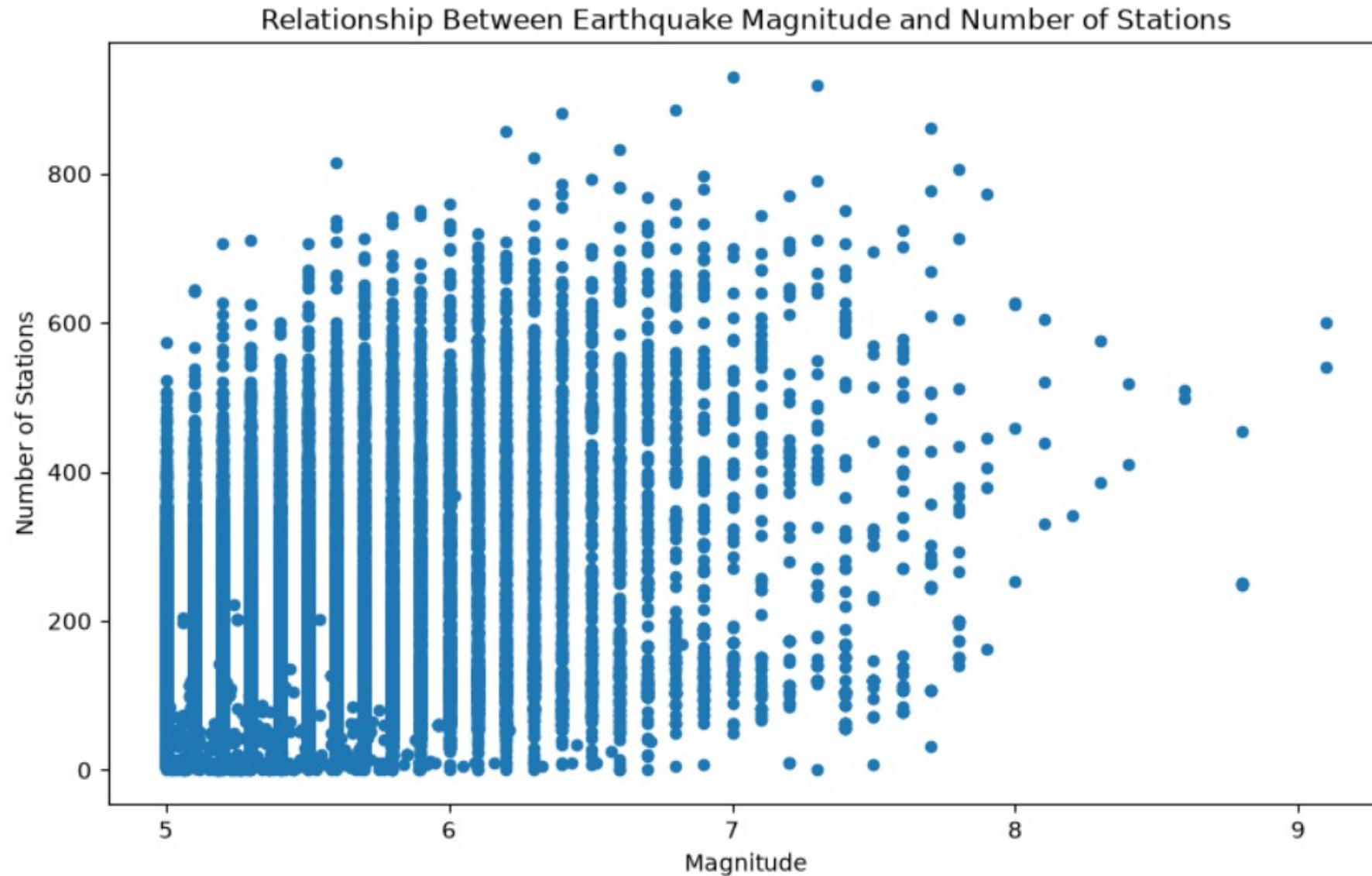
Earthquake Frequency by Time of Day (UTC)



Objective 5 - Key Findings

- Monthly frequency is fairly even: March is highest at 9,833 and February is lowest at 8,333.
- In UTC, the 0-6 period has the most records and 18-24 has the fewest.
- UTC is not local time at each earthquake location, so the time-of-day result should not be treated as a local night/day effect.

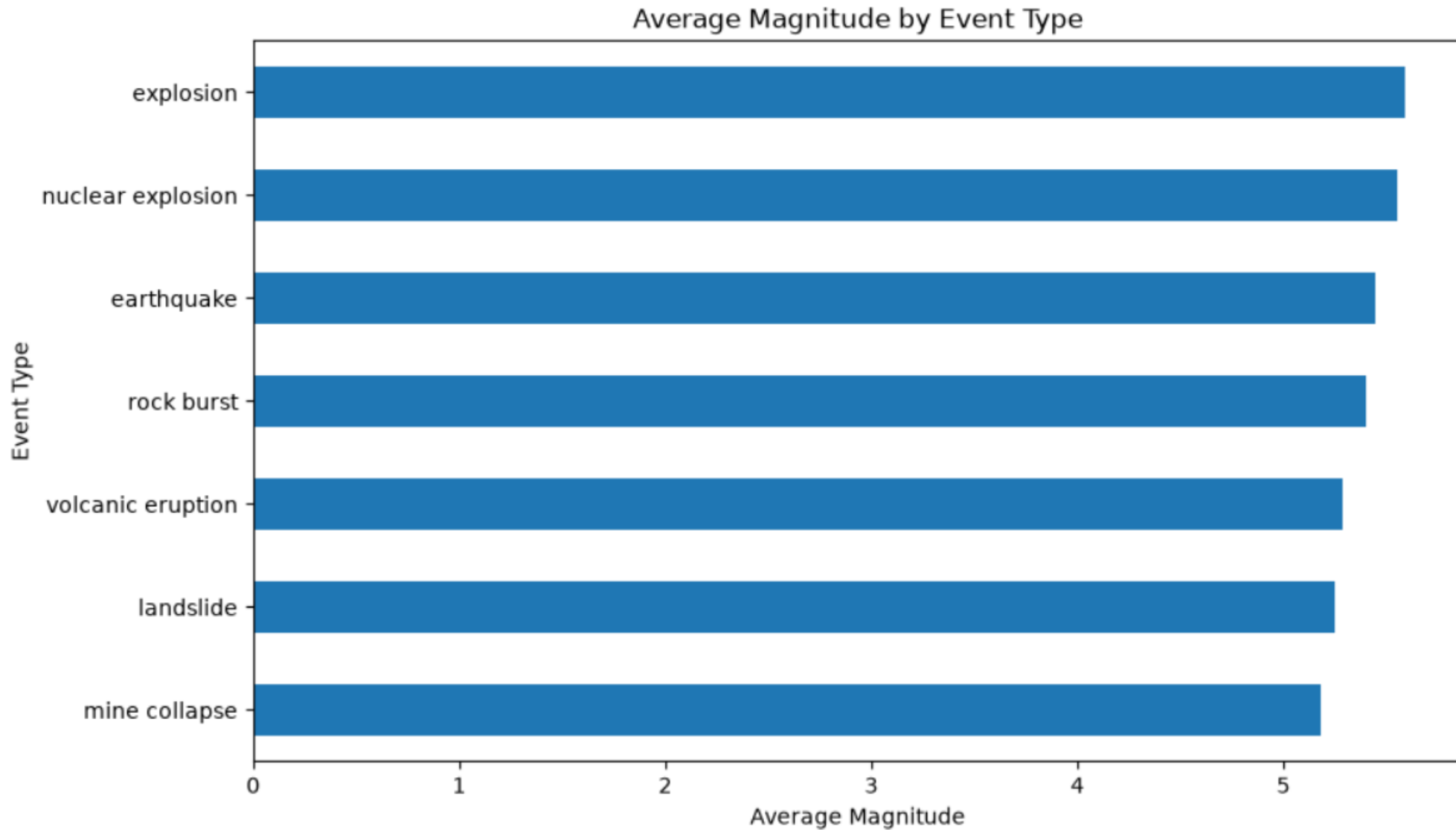
Earthquake Magnitude vs. Number of Reporting Stations



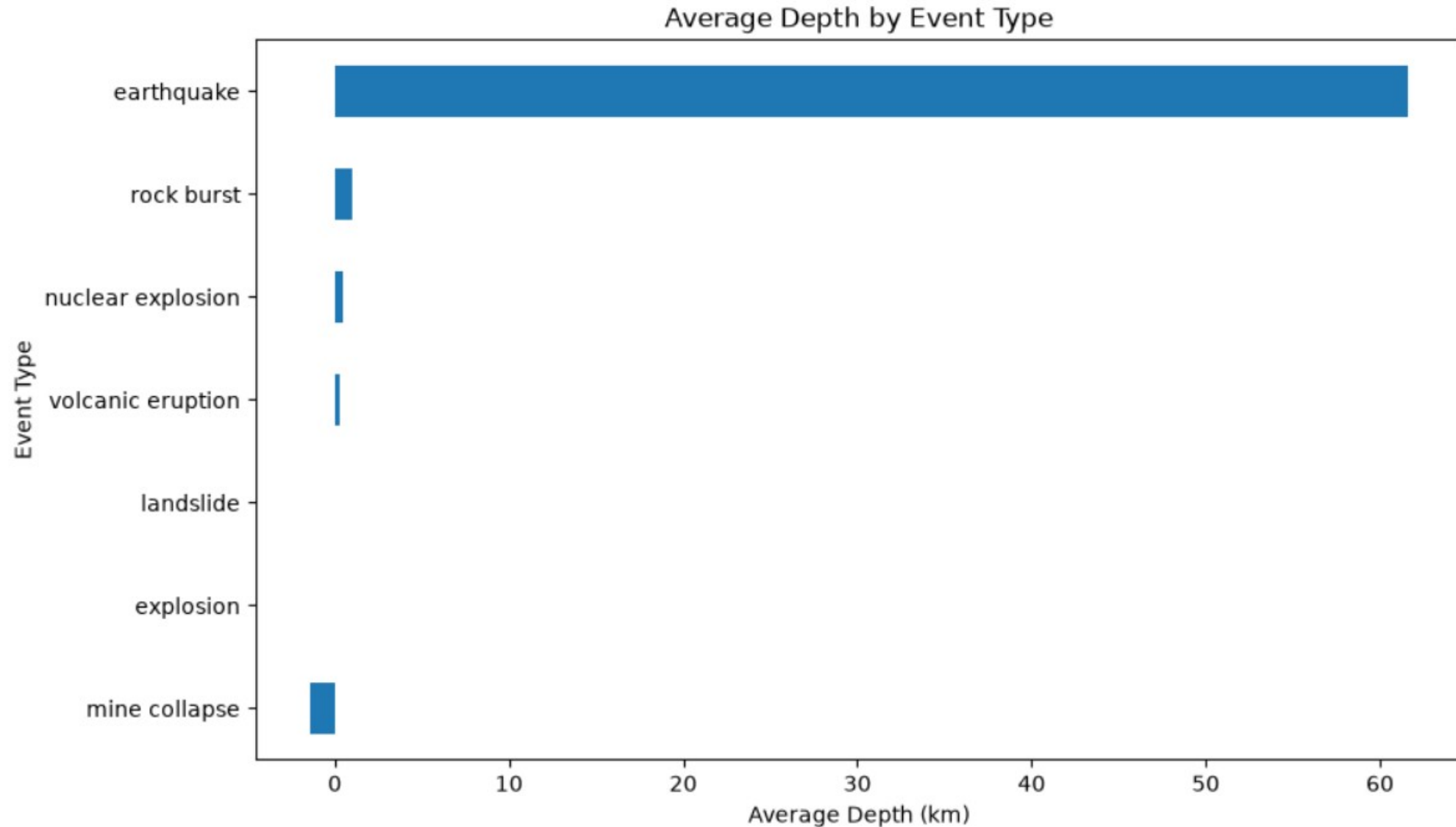
Objective 6 - Key Finding

- Magnitude and number of reporting stations have a moderate positive correlation of about 0.46.
- Higher-magnitude earthquakes tend to be recorded using more stations.
- Rows with missing nst values were excluded from this analysis.

Average Magnitude by Event Type



Average Depth by Event Type



Objective 7 - Key Findings

- Seven seismic event types appear in the dataset.
- Average magnitudes are fairly close across event types, roughly 5.18-5.59.
- Earthquakes are much deeper on average: about 61.58 km, while most other event types are near the surface.
- Some non-earthquake types have very few records, so their averages should be interpreted carefully.

Conclusion

- Historical earthquake activity is concentrated in specific locations, while magnitude patterns differ less by month or depth.
- Frequency alone is not enough to describe earthquake risk; magnitude, depth, location, exposure, and infrastructure also matter.
- Historical patterns can support preparedness, but they should not be used alone to predict future earthquakes.

Next Steps

- Combine earthquake records with population and infrastructure data.
- Add geological or tectonic-plate information for stronger location-based risk analysis.
- Convert UTC timestamps to local time before making local time-of-day comparisons.
- Continue monitoring newer data as the earthquake catalog is updated.